

ASSINIBOIA, SK, Canada

WMO: 714870

Lat: 49.730N

Lon: 105.950W

Elev: 2380

StdP: 13.48

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-17.7	-29.6	1.1	-22.9	-23.6	1.6	-17.5	31.8	18.9	29.0	18.7	8.5	270	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	89.6	64.0	85.8	62.8	82.2	61.8	68.4	80.8	66.1	78.7	64.1	77.2	10.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	99.0	73.3	61.8	90.2	71.1	59.6	83.2	68.7	34.2	80.5	32.1	79.0	30.6	77.4	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.9	24.7	21.8	-29.3	96.4	5.7	3.5	-33.4	98.9	-36.7	100.9	-39.9	102.8	-44.0	105.3		
			-29.6	72.8	5.6	3.0	-33.6	74.9	-36.9	76.7	-40.0	78.4	-44.1	80.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	11.6	13.7	25.4	40.1	51.6	60.4	66.5	65.3	55.6	41.5	26.7	15.0	
	DBStd	22.59	15.38	14.60	14.44	10.09	8.46	6.60	5.88	6.71	8.60	9.53	12.52	14.27	
	HDD50	5490	1191	1015	764	317	83	5	0	0	40	289	700	1086	
	HDD65	9509	1656	1435	1228	746	420	167	51	79	300	727	1149	1551	
	CDD50	1694	0	0	1	21	133	317	512	475	207	27	1	0	
	CDD65	240	0	0	0	0	5	29	99	89	18	0	0	0	
	CDH74	3541	0	0	0	13	151	405	1253	1309	393	17	0	0	
	CDH80	1242	0	0	0	1	30	107	449	520	134	1	0	0	
Wind	WSAvg	10.5	11.4	10.6	11.1	11.3	11.4	10.3	8.8	8.8	9.6	10.7	11.0	11.0	
Precipitation	PrecAvg	15.7	0.6	0.4	1.0	1.0	2.3	2.6	2.0	1.7	1.5	0.8	0.6	0.9	
	PrecMax	21.1	1.7	1.1	2.2	4.1	6.4	6.7	6.7	6.8	4.5	2.6	1.1	2.2	
	PrecMin	12.8	0.2	0.1	0.0	0.0	0.0	0.3	0.0	0.2	0.0	0.0	0.0	0.2	
	PrecStd	2.4	0.4	0.3	0.6	1.0	1.6	1.5	1.6	1.6	1.2	0.7	0.4	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	46.1	62.5	75.5	84.1	88.6	94.8	95.3	90.0	76.4	62.0	45.5	
		MCWB	37.0	39.6	47.5	52.4	58.1	63.6	65.8	63.9	61.8	54.7	48.2	37.8	
	2%	DB	38.6	40.0	53.7	68.0	78.3	82.8	89.1	90.3	84.1	68.1	53.1	41.1	
		MCWB	34.7	35.6	43.0	49.8	55.8	63.0	66.0	62.6	58.7	51.6	42.7	35.6	
	5%	DB	36.2	36.5	48.1	63.0	73.4	78.4	85.0	85.9	78.3	61.7	47.0	37.5	
		MCWB	33.2	33.5	40.6	47.7	53.9	60.3	65.3	61.9	57.4	48.5	39.5	33.7	
	10%	DB	33.1	33.0	42.7	57.6	68.7	74.6	80.9	81.8	72.5	56.5	42.2	33.9	
		MCWB	30.8	30.9	37.7	44.8	51.7	58.5	63.7	61.1	55.0	46.0	36.8	31.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	40.0	48.6	54.4	60.8	69.3	73.2	69.3	64.3	56.6	48.2	39.1
			MCDB	41.7	45.5	60.5	71.0	77.1	81.5	84.7	82.5	79.7	71.2	61.5	43.7
		2%	WB	35.1	36.0	44.1	51.1	57.5	65.2	69.9	66.5	61.1	52.6	43.5	36.1
			MCDB	38.1	39.2	52.1	65.4	72.3	77.3	81.8	80.4	77.9	64.7	51.7	39.8
5%		WB	33.4	33.6	40.9	48.5	55.6	62.6	67.3	64.6	58.5	49.5	40.0	33.8	
		MCDB	36.1	36.3	47.5	61.0	69.3	73.1	79.6	79.2	73.6	59.9	46.1	37.7	
10%		WB	30.7	31.0	37.6	45.8	53.7	60.4	65.3	62.8	56.2	46.9	37.0	31.1	
		MCDB	33.2	33.2	42.4	56.0	65.4	70.8	77.5	77.2	70.3	55.5	41.7	34.1	
Mean Daily Temperature Range	5% DB	MDBR	17.7	18.1	18.3	22.0	23.3	21.0	24.2	26.1	24.7	21.1	18.0	16.8	
		MCDBR	18.5	19.4	24.3	31.2	31.1	28.0	30.3	33.6	34.6	30.2	24.5	18.9	
		MCWBR	15.5	15.5	15.7	16.2	13.6	12.1	12.7	12.4	14.5	16.1	15.6	14.7	
	5% WB	MCDBR	18.1	18.7	23.5	28.4	26.3	23.5	25.4	27.1	29.8	27.6	23.5	18.6	
		MCWBR	15.4	15.3	15.6	15.6	12.7	11.8	12.6	11.8	13.6	15.4	15.5	14.9	
Clear-Sky Solar Irradiance	taub		0.251	0.256	0.271	0.324	0.345	0.344	0.357	0.365	0.323	0.281	0.253	0.230	
	taud		2.402	2.439	2.529	2.399	2.385	2.438	2.414	2.365	2.451	2.555	2.521	2.472	
	Ebn at Noon		261	289	303	293	290	290	284	276	278	276	261	258	
	Edh at Noon		19	24	27	34	36	35	35	35	29	22	17	16	
All-Sky Solar Radiation	RadAvg		413	740	1149	1541	1846	1966	2135	1717	1262	777	437	315	
	RadStd		43	54	76	104	138	120	123	121	115	89	53	28	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-1.39	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page